

06/26/25 - Video 1.9 - Properties of Asymptotic Notations

General Properties

- if $f(n)$ is $O(g(n))$ then $a \cdot f(n)$ is $O(g(n))$

↳ meaning if $f(n) = 2n^2 + 5$ is $O(n^2)$ then $7 \cdot f(n)$

$$7(2n^2 + 5)$$

$14n^2 + 35$ is also $O(n^2)$

Reflexive Properties

- if $f(n)$ is given, then $f(n)$ is $O(n^2)$

↳ meaning if $f(n) = n^2$, then $O(n^2)$

Transitive Properties

- if $f(n)$ is $O(g(n))$ and $g(n)$ is $O(h(n))$, then $f(n) = O(h(n))$

↳ very simple thing to understand here

Symmetric Properties

- if $f(n)$ is $\Theta(g(n))$ then $g(n)$ is $\Theta(f(n))$

this is **ONLY TRUE** for theta notation.

Transpose Symmetric

• if $f(n)$ is $O(g(n))$ then $g(n) = \Omega(f(n))$

↳ if $f(n) = n$ and $g(n) = n^2$
then $O(n^2)$ and $\Omega(n)$

• if $f(n) = O(g(n))$

and $d(n) = O(e(n))$

then $f(n) + d(n) = O(\max(g(n), e(n)))$

• if $f(n) = O(g(n))$

and $d(n) = O(e(n))$

then $f(n) * d(n) = O(g(n) * e(n))$

And with this piece of information,
we can say

$$f(n) = \Theta(g(n))$$

Typically when we have the upper/lower
bound, we can represent it by theta Θ .

n^2	<	n^3
$2^2 = 4$		$2^3 = 8$
$3^2 = 9$		$3^3 = 27$
$4^2 = 16$		$4^3 = 64$

we have sampled values here
and this is one way to see
which function is greater.
 $n^3 > n^2$.

n^2	<	n^3
$\log n^2$		$\log n^3$
$2 \log n$		$3 \log n$

the second method is to
apply log to both sides
and then compare. Clearly
 $3 \log n > 2 \log n$.

★ Log Formulas to Know ★

$$\log ab = \log a + \log b$$

$$\log a/b = \log a - \log b$$

$$\log a^b = b \log a$$

$$a \log^b c = b \log^a c$$

$$a^b = n \text{ then } b = \log^n a$$

$$f(n) = n^2 \log n \quad g(n) = n(\log n)^{10}$$

$$\log [n^2 \log n]$$

$$\log [n(\log n)^{10}]$$

$$\log n^2 + \log \log n$$

$$\log n + \log(\log n)^{10}$$

$$\underline{2 \log n + \log \log n}$$

$$\underline{\log n + 10 \log \log n}$$

- 2 is greater than 1
- $10 \log \log n$ is much smaller than $\log \log n$

$f(n) = n^2 \log n$ is greater

06/26/25- Video 1.10.1 Cont

$$f(n) = \frac{3n^{\sqrt{n}}}{\sqrt{n} \cdot 3n}$$

$$g(n) = \frac{2^{\sqrt{n}} \log n}{\sqrt{n} \cdot 2 \log n}$$

let's say we ignore the $\sqrt{n}$ for now. $3n$ is greater than $2 \log n$.
And always will be, we can say $f(n) > g(n)$.

$f(n) = 3n^{\sqrt{n}}$ is greater.

$$f(n) = n^{\log n}$$

$$\log n \log n$$

$$\log n \log n$$

$$g(n) = 2^{\sqrt{n}}$$

$$\log 2^{\sqrt{n}}$$

$$\sqrt{n} \log 2$$

$\log n \log n$ is a seriously small value, so $g(n)$ is greater

$g(n) = 2^{\sqrt{n}}$ is greater.

$$f(n) = 2^{\log n}$$

$$\log 2^{\log n}$$

$$\log n \log 2$$

$$g(n) = n^{\sqrt{n}}$$

$$\log n^{\sqrt{n}}$$

$$\sqrt{n} \log n$$

two logs compared to one log, $g(n)$ is greater

$g(n) = n^{\sqrt{n}}$ is greater

Log Formulas to Know

$$\log ab = \log a + \log b$$

$$\log a/b = \log a - \log b$$

$$\log a^b = b \log a$$

$$a^{\log b} = b^{\log a}$$

$$a^b = n \text{ then } b = \log^n a$$